

Exam 1 Questions (1 – 5.C)
CHEM 1310 | Spring 2026

QUESTION 1 (6 points)

Which of the following is the correct symbol for an atom or ion containing 45 protons, 58 neutrons, and 42 electrons?

- A. ☐ $^{103}\text{Rh}^{3+}$
 - B. ☐ $^{97}\text{Rh}^{3+}$
 - C. ☐ $^{58}\text{Rh}^{3+}$
 - D. ☐ $^{103}\text{Rh}^{3-}$
 - E. ☐ $^{97}\text{Rh}^{3-}$
-

Which of the following is the correct symbol for an atom or ion containing 74 protons, 108 neutrons, and 70 electrons?

- A. ☐ $^{182}\text{W}^{4+}$
 - B. ☐ $^{144}\text{W}^{4+}$
 - C. ☐ $^{108}\text{W}^{4+}$
 - D. ☐ $^{182}\text{W}^{4-}$
 - E. ☐ $^{108}\text{W}^{4-}$
-

Which of the following is the correct symbol for an atom or ion containing 40 protons, 50 neutrons, and 38 electrons?

- A. ☐ $^{90}\text{Zr}^{2+}$
- B. ☐ $^{78}\text{Zr}^{2+}$
- C. ☐ $^{50}\text{Zr}^{2+}$
- D. ☐ $^{90}\text{Zr}^{2-}$
- E. ☐ $^{50}\text{Zr}^{2-}$

QUESTION 2 (6 points)

Enter the correct element symbol (e.g., "He") based on each of the following descriptions.

The noble gas in the fifth period	Xe
The transition metal with atomic number 44	Ru
The lanthanide with the smallest atomic number	La
The alkaline earth metal in the third period	Mg

Enter the correct element symbol (e.g., "He") based on each of the following descriptions.

The noble gas in the fourth period	Kr
The transition metal with atomic number 42	Mo
The actinide with the smallest atomic number	Ac
The alkaline earth metal in the fourth period	Ca

Enter the correct element symbol (e.g., "He") based on each of the following descriptions.

The noble gas in the first period	He
The transition metal with atomic number 46	Pd
The lanthanide with the greatest atomic number	Lu
The alkaline earth metal in the second period	Be

QUESTION 3 (5 points)

A hypothetical element has only two stable isotopes, A and B. The atomic masses of the isotopes are listed below. What is the average atomic mass in atomic mass units (amu) of the hypothetical element if 71,222 of every 100,000 atoms are atoms of isotope A? Enter your response to the nearest 0.001 amu.

Isotope A = 49.00 amu

Isotope B = 47.00 amu

48.42 amu

A hypothetical element has only two stable isotopes, A and B. The atomic masses of the isotopes are listed below. What is the average atomic mass in atomic mass units (amu) of the hypothetical element if 71,222 of every 100,000 atoms are atoms of isotope B? Enter your response to the nearest 0.01 amu.

Isotope A = 49.00 amu

Isotope B = 47.00 amu

47.58 amu

A hypothetical element has only two stable isotopes, A and B. The atomic masses of the isotopes are listed below. What is the average atomic mass in atomic mass units (amu) of the hypothetical element if 45,438 of every 100,000 atoms are atoms of isotope A? Enter your response to the nearest 0.01 amu.

Isotope A = 49.00 amu

Isotope B = 47.00 amu

47.91 amu

QUESTION 4 (6 points)

The compound retinol has the formula $C_{20}H_{30}O$. Indicate whether each statement below regarding this compound is true or false.

[☐ **True** ☐ False] The molar mass of this compound is 286.5 g/mol.

[☐ True ☐ **False**] Each molecule of retinol in a bulk sample has an identical mass of 286.5 g/mol.

[☐ True ☐ **False**] Although the average number of carbon atoms per molecule is 20, retinol molecules contain a distribution of different numbers of carbon atoms.

[☐ **True** ☐ False] In a sample of retinol weighing 286.5 g, there are 6.022×10^{23} oxygen atoms.

The compound retinol has the formula $C_{20}H_{30}O$. Indicate whether each statement below regarding this compound is true or false.

- [☐ **True** ☐ False] The molar mass of this compound is 286.5 g/mol.
[☐ **True** ☐ False] The average mass of a retinol molecule in a bulk sample of this compound is 286.5 amu.
[☐ True ☐ **False**] Although the average number of hydrogen atoms per molecule is 30, retinol molecules contain a distribution of different numbers of hydrogen atoms.
[☐ True ☐ **False**] In a sample of retinol weighing 286.5 g, there are 6.022×10^{23} carbon atoms.
-

The compound retinol has the formula $C_{20}H_{30}O$. Indicate whether each statement below regarding this compound is true or false.

- [☐ True ☐ **False**] Two moles of this compound have a mass of 286.5 grams.
[☐ **True** ☐ False] The average mass of a retinol molecule in a bulk sample of this compound is 286.5 amu.
[☐ True ☐ **False**] Although the average number of hydrogen atoms per molecule is 30, retinol molecules contain a distribution of different numbers of hydrogen atoms.
[☐ True ☐ **False**] In a sample of retinol weighing 286.5 g, there are 6.022×10^{23} hydrogen atoms.

QUESTION 5 (6 points)

Indicate whether each of the following chemical names corresponds correctly to the given formula.

- | | |
|-------------------------------------|--|
| S_4N_4 tetrasulfur tetranitride | [☐ Correct ☐ Incorrect] |
| SO_3 sulfur(IV) oxide | [☐ Correct ☐ Incorrect] |
| $CaSO_3$ calcium sulfate | [☐ Correct ☐ Incorrect] |
| $Fe_3(PO_4)_2$ iron(II) phosphate | [☐ Correct ☐ Incorrect] |
-

Indicate whether each of the following chemical names corresponds correctly to the given formula.

S_4N_4 | sulfur tetranitride [☐ Correct ☒ **Incorrect**]

SO_3 | sulfur trioxide [☒ **Correct** ☐ Incorrect]

CaSO_3 | calcium sulfite [☒ **Correct** ☐ Incorrect]

$\text{Fe}_3(\text{PO}_4)_2$ | iron(III) phosphate [☐ Correct ☒ **Incorrect**]

Indicate whether each of the following chemical names corresponds correctly to the given formula.

S_4N_4 | tetrasulfur tetranitride [☒ **Correct** ☐ Incorrect]

SO_3 | sulfur(III) oxide [☐ Correct ☒ **Incorrect**]

CaSO_3 | calcium sulfite [☒ **Correct** ☐ Incorrect]

$\text{Fe}_3(\text{PO}_4)_2$ | triiron diphosphate [☐ Correct ☒ **Incorrect**]

QUESTION 6 (6 points)

How many moles of **sodium** are present in a sample of Na_2CrO_4 weighing 310.2 g? The molar mass of Na_2CrO_4 is 161.97 g/mol. Enter your response in moles (mol) to the nearest 0.01 mol.

3.83 mol

How many moles of **hydrogen** are present in a sample of $(\text{NH}_4)_2\text{CrO}_4$ weighing 310.2 g? The molar mass of $(\text{NH}_4)_2\text{CrO}_4$ is 152.07 g/mol. Enter your response in moles (mol) to the nearest 0.01 mol.

16.32 mol

How many moles of **chromium** are present in a sample of $\text{Na}_2\text{Cr}_2\text{O}_7$ weighing 510.2 g? The molar mass of $\text{Na}_2\text{Cr}_2\text{O}_7$ is 209.97 g/mol. Enter your response in moles (mol) to the nearest 0.01 mol.

4.86 mol

QUESTION 7 (6 points)

In a compound containing only nitrogen and oxygen, the percent by mass of nitrogen is 25.94%. What is the empirical formula of this compound?

- A. ☐ N_2O_5
 - B. ☐ N_2O
 - C. ☐ NO_2
 - D. ☐ NO
 - E. ☐ N_2O_3
-

In a compound containing only nitrogen and oxygen, the percent by mass of nitrogen is 46.68%. What is the empirical formula of this compound?

- A. ☐ N_2O_5
 - B. ☐ N_2O
 - C. ☐ NO_2
 - D. ☒ NO
 - E. ☐ N_2O_3
-

In a compound containing only nitrogen and oxygen, the percent by mass of nitrogen is 30.45%. What is the empirical formula of this compound?

- A. ☐ N_2O_5
- B. ☐ N_2O
- C. ☒ NO_2
- D. ☐ NO
- E. ☐ N_2O_3

QUESTION 8 (6 points)

Consider what happens when aqueous calcium nitrate and aqueous sodium carbonate are mixed. Using the dropdown menus below, indicate the identity of the solid product that forms (if any) and the molar ratio of calcium nitrate to sodium carbonate in the balanced chemical equation.

Solid product: [☒ **Calcium carbonate** ☐ Sodium nitrate ☐ None]

Calcium nitrate/sodium carbonate ratio: [☒ **1:1** ☐ 1:2 ☐ 2:1 ☐ 2:3 ☐ 3:2]

Consider what happens when aqueous lead(II) nitrate and aqueous potassium iodide are mixed. Using the dropdown menus below, indicate the identity of the solid product that forms (if any) and the molar ratio of lead(II) nitrate to potassium iodide in the balanced chemical equation.

Solid product: [☒ **Lead(II) iodide** ☐ Potassium nitrate ☐ None]

Lead(II) nitrate/potassium iodide ratio: [☐ 1:1 ☒ **1:2** ☐ 2:1 ☐ 2:3 ☐ 3:2]

Consider what happens when aqueous barium chloride and aqueous sodium sulfate are mixed. Using the dropdown menus below, indicate the identity of the solid product that forms (if any) and the molar ratio of barium chloride to sodium sulfate in the balanced chemical equation.

Solid product: [☒ **Barium sulfate** ☐ Sodium chloride ☐ None]

Barium chloride/sodium sulfate ratio: [☒ **1:1** ☐ 1:2 ☐ 2:1 ☐ 2:3 ☐ 3:2]

QUESTION 9 (6 points)

Indicate whether each solute below forms a strong electrolyte, weak electrolyte, or nonelectrolyte solution when dissolved in water.

Acetic acid (HO_2CCH_3)

[☐ strong electrolyte ☒ **weak electrolyte** ☐ nonelectrolyte]

1-Propanol ($\text{C}_3\text{H}_7\text{OH}$)

[☐ strong electrolyte ☐ weak electrolyte ☒ **nonelectrolyte**]

Potassium perchlorate (KClO_4)

[☒ **strong electrolyte** ☐ weak electrolyte ☐ nonelectrolyte]

Indicate whether each solute below forms a strong electrolyte, weak electrolyte, or nonelectrolyte solution when dissolved in water.

2-Propanol ($\text{C}_3\text{H}_7\text{OH}$)

[☐ strong electrolyte ☐ weak electrolyte ☒ **nonelectrolyte**]

Butanoic acid ($\text{HO}_2\text{CC}_3\text{H}_7$)

[☐ strong electrolyte ☒ **weak electrolyte** ☐ nonelectrolyte]

Sodium chlorate (NaClO_3)

[☒ **strong electrolyte** ☐ weak electrolyte ☐ nonelectrolyte]

Indicate whether each solute below forms a strong electrolyte, weak electrolyte, or nonelectrolyte solution when dissolved in water.

Sodium bromide (NaBr)

[☒ **strong electrolyte** ☐ weak electrolyte ☐ nonelectrolyte]

Phosphoric acid (H_3PO_4)

[☐ strong electrolyte ☒ **weak electrolyte** ☐ nonelectrolyte]

Diethyl ether ($\text{C}_4\text{H}_{10}\text{O}$)

[☐ strong electrolyte ☐ weak electrolyte ☒ **nonelectrolyte**]

QUESTION 10 (5 points)

What mass of calcium nitrate ($\text{Ca}(\text{NO}_3)_2$, 164.09 g/mol) is needed to prepare 4.25 L of a solution that is 0.275 M in **nitrate ions**? Enter your response in grams (g) to the nearest 0.1 g.

95.9 g

What mass of sodium sulfate (Na_2SO_4 , 142.04 g/mol) is needed to prepare 3.25 L of a solution that is 0.075 M in **sodium ions**? Enter your response in grams (g) to the nearest 0.1 g.

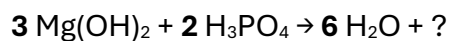
17.3 g

What mass of magnesium chloride (MgCl_2 , 95.2 g/mol) is needed to prepare 6.25 L of a solution that is 0.100 M in **chloride ions**? Enter your response in grams (g) to the nearest 0.1 g.

29.8 g

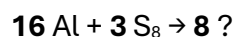
QUESTION 11 (5 points)

In an experiment involving the combination of magnesium hydroxide and phosphoric acid, the stoichiometry in bold below was observed. What is the chemical formula of the missing product?



- A. ☐ $\text{Mg}_3(\text{PO}_4)_2$
 - B. ☐ $\text{Mg}_2(\text{PO}_4)_3$
 - C. ☐ MgPO_4
 - D. ☐ Mg_3PO_4
 - E. ☐ $\text{Mg}(\text{PO}_4)_2$
-

In an experiment involving the combination of aluminum and sulfur, the stoichiometry in bold below was observed. What is the chemical formula of the missing product?



- A. ☐ **Al₂S₃**
 - B. ☐ Al₃S₂
 - C. ☐ AlS
 - D. ☐ Al₃S
 - E. ☐ AlS₂
-

In an experiment involving the combination of calcium chloride and sodium phosphate, the stoichiometry in bold below was observed. What is the chemical formula of the missing product?



- A. ☐ **Ca₃(PO₄)₂**
- B. ☐ Ca₂(PO₄)₃
- C. ☐ CaPO₄
- D. ☐ Ca₃PO₄
- E. ☐ Ca(PO₄)₂

QUESTION 12 (5 points)

What aqueous species are produced, and in what amounts, when 1 mole of NaNO₃ is dissolved in water?

- A. ☐ **1 mol Na⁺(aq) and 1 mol NO₃⁻(aq)**
 - B. ☐ 1 mol Na⁺, 1 mol N⁵⁺(aq), and 3 mol O²⁻(aq)
 - C. ☐ 1 mol Na⁺(aq), 0.5 mol N₂(aq), and 1.5 mol O₂(aq)
 - D. ☐ 1 mol Na(aq) and 1 mol NO₃(aq)
 - E. ☐ 1 mol NaNO₃(aq); this compound does not dissociate in solution.
-

What aqueous species are produced, and in what amounts, when 1 mole of NaBrO_3 is dissolved in water?

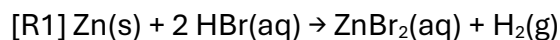
- A. ☐ 1 mol $\text{Na}^+(\text{aq})$ and 1 mol $\text{BrO}_3^-(\text{aq})$
 - B. ☐ 1 mol $\text{Na}^+(\text{aq})$, 1 mol $\text{Br}^{5+}(\text{aq})$, and 3 mol $\text{O}^{2-}(\text{aq})$
 - C. ☐ 1 mol $\text{Na}^+(\text{aq})$, 1 mol $\text{Br}^-(\text{aq})$, and 3 mol $\text{O}(\text{aq})$
 - D. ☐ 1 mol $\text{Na}^+(\text{aq})$, 1 mol $\text{Br}^-(\text{aq})$, and 1.5 mol $\text{O}_2(\text{aq})$
 - E. ☐ 1 mol $\text{NaBrO}_3(\text{aq})$; this compound does not dissociate in solution.
-

What aqueous species are produced, and in what amounts, when 1 mole of NaClO_4 is dissolved in water?

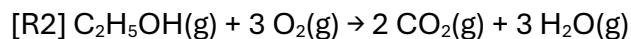
- A. ☐ 1 mol $\text{Na}^+(\text{aq})$ and 1 mol $\text{ClO}_4^-(\text{aq})$
- B. ☐ 1 mol $\text{Na}(\text{aq})$ and 1 mol $\text{ClO}_4(\text{aq})$
- C. ☐ 1 mol $\text{Na}^+(\text{aq})$, 1 mol $\text{Cl}^{7+}(\text{aq})$, and 4 mol $\text{O}^{2-}(\text{aq})$
- D. ☐ 1 mol $\text{Na}^+(\text{aq})$ and 1 mol $\text{ClO}_4(\text{aq})$
- E. ☐ 1 mol $\text{NaClO}_4(\text{aq})$; this compound does not dissociate in solution.

QUESTION 13 (6 points)

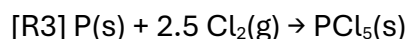
Classify each of the following reactions as synthesis, decomposition, single replacement, double replacement, or combustion.



single replacement

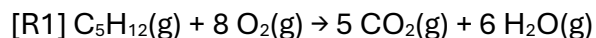


combustion

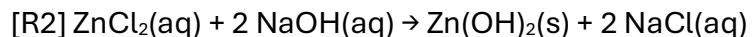


synthesis

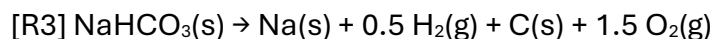
Classify each of the following reactions as synthesis, decomposition, single replacement, double replacement, or combustion.



combustion

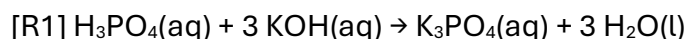


double replacement

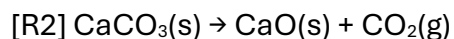


decomposition

Classify each of the following reactions as synthesis, decomposition, single replacement, double replacement, or combustion.



double replacement



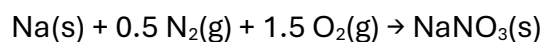
decomposition



synthesis

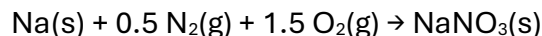
QUESTION 14 (5 points)

How many moles of oxygen gas are needed to prepare 8.40 moles of sodium nitrate according to the balanced chemical equation below? Assume excess amounts of all other reactants.



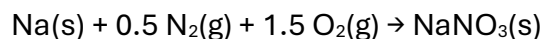
- A. ☐ 12.6 mol
 - B. ☐ 8.40 mol
 - C. ☐ 5.60 mol
 - D. ☐ 4.20 mol
 - E. ☐ 16.8 mol
-

How many moles of nitrogen gas are needed to prepare 8.40 moles of sodium nitrate according to the balanced chemical equation below? Assume excess amounts of all other reactants.



- A. ☐ 12.6 mol
 - B. ☐ 8.40 mol
 - C. ☐ 5.60 mol
 - D. ☒ 4.20 mol
 - E. ☐ 16.8 mol
-

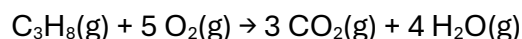
How many moles of sodium nitrate are produced when 8.60 moles of oxygen gas are reacted with excess sodium and nitrogen according to the balanced chemical equation below? Assume 100% yield.



- A. ☒ 5.73 mol
- B. ☐ 8.60 mol
- C. ☐ 12.9 mol
- D. ☐ 4.30 mol
- E. ☐ 17.2 mol

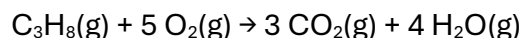
QUESTION 15 (5 points)

What mass of carbon dioxide (CO_2 , 44.0 g/mol) is produced when 10.5 kg of propane (C_3H_8 , 44.1 g/mol) is fully combusted according to the balanced reaction below? Assume that the reaction goes to completion. Enter your response in kilograms (kg) to the nearest 0.1 kg.



31.4 kg

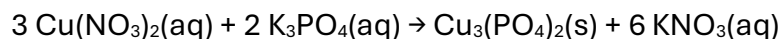
What mass of water (H_2O , 18.0 g/mol) is produced when 10.5 kg of propane (C_3H_8 , 44.1 g/mol) is fully combusted according to the balanced reaction below? Assume that the reaction goes to completion. Enter your response in kilograms (kg) to the nearest 0.1 kg.



17.1 kg

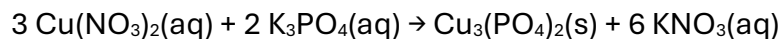
QUESTION 16 (5 points)

Masses of 0.132 g **each** of $\text{Cu}(\text{NO}_3)_2$ (187.55 g/mol) and K_3PO_4 (212.27 g/mol) are mixed in aqueous solution and react according to the balanced chemical equation below. Solid $\text{Cu}_3(\text{PO}_4)_2$ (380.58 g/mol) with a mass of 0.095 g was obtained. What percent yield is implied by this result?



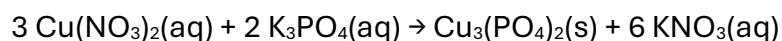
- A. ☐ 106%
 - B. ☐ 68%
 - C. ☐ 94%
 - D. ☐ 0.53%
 - E. ☐ 112%
-

Masses of 0.187 g **each** of $\text{Cu}(\text{NO}_3)_2$ (187.55 g/mol) and K_3PO_4 (212.27 g/mol) are mixed in aqueous solution and react according to the balanced chemical equation below. Solid $\text{Cu}_3(\text{PO}_4)_2$ (380.58 g/mol) with a mass of 0.095 g was obtained. What percent yield is implied by this result?



- A. ☐ 75%
 - B. ☐ 63%
 - C. ☐ 101%
 - D. ☐ 56%
 - E. ☐ 86%
-

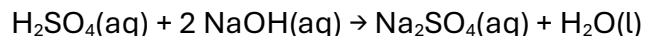
Masses of 0.200 g **each** of $\text{Cu}(\text{NO}_3)_2$ (187.55 g/mol) and K_3PO_4 (212.27 g/mol) are mixed in aqueous solution and react according to the balanced chemical equation below. Solid $\text{Cu}_3(\text{PO}_4)_2$ (380.58 g/mol) with a mass of 0.095 g was obtained. What percent yield is implied by this result?



- A. ☐ 70%
- B. ☐ 59%
- C. ☐ 81%
- D. ☐ 95%
- E. ☐ 104%

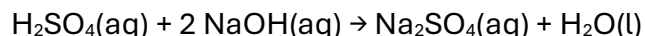
QUESTION 17 (5 points)

An aqueous solution of sulfuric acid with volume 250. mL and molarity 0.125 M was subjected to complete neutralization according to the balanced chemical equation below, using 300. mL of a sodium hydroxide solution. What is the minimum molarity of the sodium hydroxide solution required to accomplish this? Enter your response in moles per liter (mol/L, M) to the nearest 0.001 M.



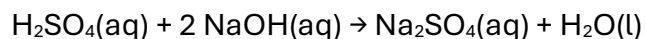
0.208 M

An aqueous solution of sulfuric acid with volume 450. mL and molarity 0.115 M was subjected to complete neutralization according to the balanced chemical equation below, using 300. mL of a sodium hydroxide solution. What is the minimum molarity of the sodium hydroxide solution required to accomplish this? Enter your response in moles per liter (mol/L, M) to the nearest 0.001 M.



0.345 M

An aqueous solution of sulfuric acid with volume 250. mL and molarity 0.115 M was subjected to complete neutralization according to the balanced chemical equation below, using 250. mL of a sodium hydroxide solution. What is the minimum molarity of the sodium hydroxide solution required to accomplish this? Enter your response in moles per liter (mol/L, M) to the nearest 0.001 M.



0.230 M

QUESTION 18 (6 points)

P₄, known as “white phosphorus,” is a polyatomic form of phosphorus. Indicate whether each statement regarding this substance is true or false.

- [☐ True ☒ **False**] The average mass of a molecule of P₄ is 30.97 amu.
[☐ **True** ☐ False] The empirical formula of white phosphorus is P.
[☐ **True** ☐ False] There are four times as many phosphorus atoms as P₄ molecules in any sample of white phosphorus.
[☐ True ☒ **False**] P₄ is best described as a compound because it contains molecules rather than atoms.
-

P₄, known as “white phosphorus,” is a polyatomic form of phosphorus. Indicate whether each statement regarding this substance is true or false.

- [☐ **True** ☐ False] The average mass of a molecule of P₄ is 123.9 amu.
[☐ **True** ☐ False] The empirical formula of white phosphorus is P.
[☐ True ☒ **False**] There are four times as many P₄ molecules as phosphorus atoms in any sample of white phosphorus.
[☐ True ☒ **False**] P₄ is best described as a compound because it contains molecules rather than atoms.
-

P_4 , known as “white phosphorus,” is a polyatomic form of phosphorus. Indicate whether each statement regarding this substance is true or false.

[☐ **True** ☐ False] The average mass of a molecule of P_4 is 123.9 amu.

[☐ True ☒ **False**] The empirical formula of white phosphorus is P_4 .

[☐ **True** ☐ False] There are four times as many phosphorus atoms as P_4 molecules in any sample of white phosphorus.

[☐ **True** ☐ False] P_4 is best described as an element because it contains only one type of atom.